Concord-SLAM: Cross-Render Concordance for Boundary-Native Semantic Gaussian SLAM

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Abstract

Three-dimensional Gaussian Splatting (3DGS) has recently become a promising representation for dense SLAM, due to its explicit scene parameterization, efficient differentiable rendering, and high-quality visual reconstruction. Extending 3DGS to semantic SLAM enables the construction of maps that are not only geometrically accurate but also semantically interpretable. However, existing semantic Gaussian SLAM methods usually optimize semantic, depth, and normal renderings through separate supervision objectives. Such a design underexplores the fact that these outputs are generated from the same Gaussian field and should therefore provide a concordant explanation of the underlying scene structure. This limitation is especially problematic around object boundaries and geometric discontinuities, where inaccurate Gaussian growth can cause cross-boundary expansion, boundary blurring, and semantic leakage. To address this problem, we propose Concord-SLAM, a boundary-native semantic Gaussian SLAM framework that establishes concordance among semantic, depth, and normal renderings while promoting boundary-aware Gaussian growth. Our framework contains two key components. First, Cross-Render Concordance (CRC) jointly examines semantic, depth, and normal views rendered from the same Gaussian map, and converts their structural discrepancies in boundary responses, regional continuity, and local geometry into optimization signals. In this way, disagreement among different renderings is used as an intrinsic cue for improving the semantic-geometric coherence of the Gaussian field. Second, Boundary-Native Splatting (BNS) augments the standard opacity- and depth-residual-based Gaussian growth rule with an additional semantic-boundary insertion mask. It actively inserts new Gaussians in under-covered neighborhoods around semantic boundaries, encouraging primitives on different sides of a boundary to be represented more separately during map construction rather than correcting boundary mixing only through later optimization. Experiments on Replica and ScanNet demonstrate improved semantic mapping and reconstruction quality, while maintaining competitive camera tracking performance.

1 Introduction

Dense SLAM estimates camera motion while reconstructing geometrically accurate and visually faithful 3D maps, and has become a fundamental capability for robot perception, augmented reality, and 3D scene understanding Sucar et al. [2021], Zhu et al. [2022], Johari et al. [2023], Sandström et al. [2023]. Semantic SLAM further augments such maps with category- or instance-level information, enabling scene-level reasoning about objects, regions, and layouts McCormac et al. [2017], Li et al. [2024a], Haghighi et al. [2023]. Recently, 3D Gaussian Splatting (3DGS) has emerged as an effective representation for online dense SLAM, owing to its explicit Gaussian primitives, efficient

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differentiable rendering, local map updates, and high-quality novel-view synthesis Kerbl et al. [2023], Yan et al. [2024], Matsuki et al. [2024], Keetha et al. [2024], Li et al. [2024d]. These properties make 3DGS particularly suitable for semantic Gaussian SLAM, where appearance, geometry, and semantics are expected to be maintained in a unified map Li et al. [2024c,b], Zhu et al. [2025], Li et al. [2025], Lu et al. [2026], Yang et al. [2025]. However, introducing semantics into Gaussian SLAM is not simply a matter of attaching semantic attributes to Gaussians or adding an extra semantic loss. Existing semantic Gaussian SLAM methods often supervise appearance, depth, semantics, and normals through separate branches or independent objectives Li et al. [2024c,b], Zhu et al. [2025], Lu et al. [2026]. Although such supervision uses multiple signals, it underexplores a key property of Gaussian SLAM: semantic maps, depth maps, and normal fields are different renderings of the same Gaussian scene representation. Therefore, they should not only match their own supervision targets, but also provide a structurally consistent explanation of the same scene.

Our key observation is that, in a faithful semantic Gaussian map, semantic transitions, depth discontinuities, and normal variations should be spatially aligned around object boundaries and surface changes, while they should vary coherently in smooth regions. When these renderings disagree in boundary locations, regional continuity, or local geometry, the disagreement often indicates defects in the Gaussian field, such as inaccurate primitive placement, cross-boundary expansion, insufficient boundary modeling, or ambiguous semantic assignment. Since this discrepancy is generated by the representation itself, it can serve as an intrinsic signal for map refinement without requiring additional annotations. This perspective suggests that semantic Gaussian SLAM requires more than conventional multimodal fusion. Feature concatenation, cross-modal interaction, or decoder-level aggregation can combine cues, but they do not explicitly verify whether different renderings of the same Gaussian map give a concordant structural explanation Li et al. [2024c,b], Zhu et al. [2025], Lu et al. [2026], Yang et al. [2025]. This issue is especially important for explicit Gaussian representations, because primitives that grow across object boundaries or mixed surfaces during early mapping may cause boundary blurring and semantic leakage that later semantic supervision cannot fully remove. Effective semantic Gaussian SLAM should therefore combine rendering-level concordance constraints with representation-level boundary-aware Gaussian growth.

Motivated by this, we propose Concord-SLAM: Cross-Render Concordance for Boundary-Native Semantic Gaussian SLAM. Concord-SLAM uses inconsistencies among semantic, depth, and normal renderings as endogenous cues for correcting the Gaussian map, while encouraging Gaussian primitives to respect physical and semantic discontinuities during map construction. It contains two complementary components. Cross-Render Concordance (CRC) models structural agreement among semantic, depth, and normal renderings, and converts cross-render discrepancies into optimization signals for semantic-geometric refinement. Boundary-Native Splatting (BNS) augments opacity- and depth-residual-based Gaussian growth with semantic-boundary-aware insertion, allowing primitives to better conform to object boundaries during map construction. Together, CRC and BNS constrain Concord-SLAM at both the rendering-optimization level and the representation-growth level, enabling semantic Gaussian maps with accurate geometry, faithful appearance, clearer boundaries, and more stable semantics. The main contributions of this paper are summarized as follows:

- We propose Concord-SLAM, a boundary-native semantic Gaussian SLAM framework that treats semantic, depth, and normal renderings as coupled evidence of a shared Gaussian map, and uses their disagreement as an intrinsic signal for representation refinement.
- We introduce Cross-Render Concordance (CRC), which models structural agreement among semantic boundaries, depth discontinuities, and normal variations, and converts cross-render discrepancies into optimization constraints.
- We design Boundary-Native Splatting (BNS), which augments opacity- and depth-residual-based Gaussian growth with semantic-boundary-aware insertion, allowing primitives to better conform to object boundaries during map construction.
- We evaluate Concord-SLAM on Replica and ScanNet, demonstrating improvements in camera tracking, scene reconstruction, and semantic mapping. The open-source code is available at *https://anonymous.4open.science/r/concordslam*.

2 Related Work

Gaussian SLAM 3D Gaussian Splatting has been increasingly adopted in SLAM due to its explicit primitive-based representation, differentiable splatting, and efficient rendering, making it suitable for camera tracking, dense mapping, and novel-view synthesis Kerbl et al. [2023], Yan et al. [2024], Matsuki et al. [2024], Keetha et al. [2024]. GS-SLAM introduces 3D Gaussians as a dense visual SLAM map representation, while Gaussian Splatting SLAM further explores Gaussian-based mapping for monocular SLAM and RGB-D SLAM Yan et al. [2024], Matsuki et al. [2024]. SplaTAM jointly optimizes camera poses and Gaussian maps for RGB-D dense SLAM, showing the effectiveness of splat-based scene representations for online tracking and mapping Keetha et al. [2024]. For monocular or RGB-only settings, MonoGS++ improves Gaussian SLAM with sparse point initialization, dynamic insertion, densification, and planar regularization, further demonstrating the potential of Gaussian representations for efficient monocular RGB SLAM Li et al. [2024d]. Compared with earlier neural dense SLAM systems based on implicit fields or point-based representations, Gaussian SLAM methods benefit from explicit primitives and efficient differentiable rendering, which are more suitable for real-time map optimization and view synthesis Sucar et al. [2021], Zhu et al. [2022], Johari et al. [2023], Sandström et al. [2023], Kerbl et al. [2023]. Overall, these methods demonstrate the effectiveness of 3DGS for real-time tracking, rendering, and geometric reconstruction, but they mainly focus on geometry and appearance, with limited attention to semantic understanding and semantic boundary modeling Yan et al. [2024], Matsuki et al. [2024], Keetha et al. [2024], Li et al. [2024d].

Semantic Gaussian SLAM Semantic Gaussian SLAM associates Gaussian primitives with semantic labels, semantic features, or category-aware representations, enabling 3D maps to encode scene semantics in addition to geometry and appearance Li et al. [2024c,b], Zhu et al. [2025], Li et al. [2025], Lu et al. [2026], Yang et al. [2025]. SGS-SLAM incorporates appearance, geometry, and semantic features into a multi-channel optimization framework, while GS^{3}LAM formulates RGB, depth, and semantic observations within a unified semantic Gaussian representation for dense semantic mapping Li et al. [2024c,b]. GS^{3}LAM further introduces semantic Gaussian splatting for semantic SLAM, while SemGauss-SLAM embeds semantic information into Gaussian primitives for unified tracking, semantic mapping, and high-quality rendering Li et al. [2024b], Zhu et al. [2025]. Hier-SLAM scales semantic Gaussian SLAM with hierarchically categorical Gaussian splatting, and GSFF-SLAM constructs a semantic Gaussian representation through feature fields to improve semantic mapping capability Li et al. [2025], Lu et al. [2026]. OpenGS-SLAM extends semantic Gaussian SLAM to open-set dense semantic mapping by combining 3D Gaussian Splatting with object-level scene understanding Yang et al. [2025]. Although these methods benefit from explicit and optimizable Gaussian maps, they usually treat color, depth, semantics, or related geometric cues as relatively separate supervision signals, leaving the structural coupling among renderings from the same Gaussian field underexplored Li et al. [2024c,b], Zhu et al. [2025], Li et al. [2025], Lu et al. [2026], Yang et al. [2025]. Moreover, existing Gaussian growth strategies often lack explicit awareness of semantic and geometric boundaries, which may cause cross-boundary expansion, blurred object borders, and semantic leakage Li et al. [2024c,b], Zhu et al. [2025], Yang et al. [2025]. In contrast, Concord-SLAM exploits concordance among semantic, depth, and normal renderings from the same Gaussian field, and introduces boundary-native Gaussian allocation to construct semantic Gaussian maps with clearer boundaries and stronger structural coherence.

3 Method

Given RGB-D semantic observations $\{I_t^*, D_t^*, Y_t^*\}_{t=1}^T$, where I_t^*, D_t^*, and Y_t^* denote RGB, depth, and semantic observations, Concord-SLAM maintains an explicit semantic Gaussian map $\mathcal{G}$ for geometry, appearance, and semantics. To this end, we introduce two complementary mechanisms. Cross-Render Concordance (CRC) uses rendered depth and depth-derived normals as structural evidence to correct semantic prediction. Boundary-Native Splatting (BNS) inserts additional Gaussians near under-covered semantic boundaries, encouraging the two sides of a boundary to be represented by separable primitives during map growth. The overall pipeline of Concord-SLAM is illustrated in Figure 1.

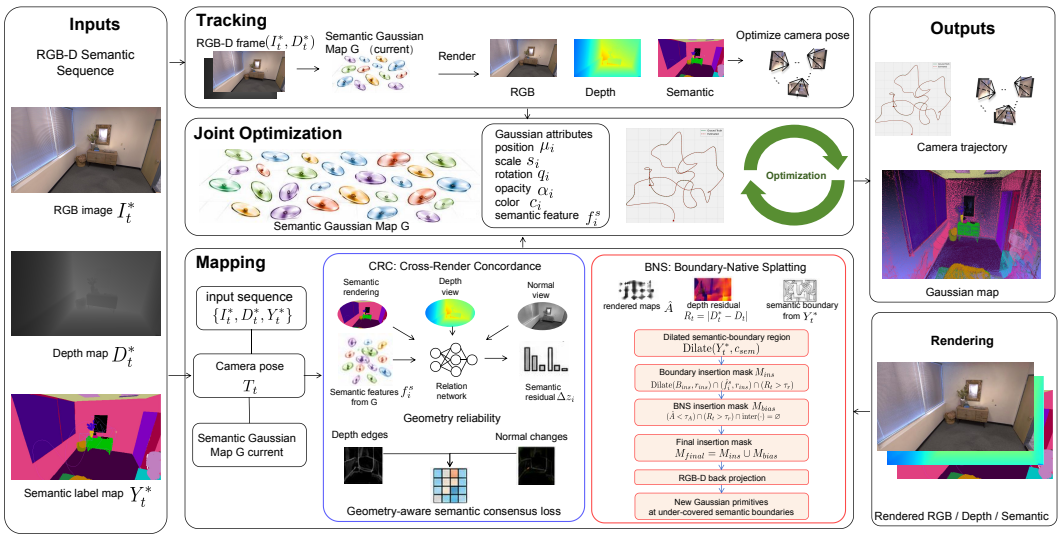


Figure 1: Overall pipeline of Concord-SLAM. The system maintains a semantic Gaussian map and jointly performs tracking, mapping, and rendering. CRC enforces cross-render concordance among semantic, depth, and normal outputs, while BNS inserts boundary-native Gaussians to improve semantic boundary representation.

3.1 Semantic Gaussian Map Representation

Concord-SLAM represents the scene as a set of optimizable semantic Gaussian primitives:

$$\mathcal{G} = \{g_i\}_{i=1}^{N}, \quad g_i = \{\boldsymbol{\mu}_i, \mathbf{s}_i, \mathbf{q}_i, \alpha_i, \mathbf{c}_i, \mathbf{f}_i^s\}. \tag{1}$$

Here, $\boldsymbol{\mu}_i$ denotes the Gaussian center, $\mathbf{s}_i$ denotes the scale, $\mathbf{q}_i$ denotes the rotation, α_i denotes the opacity, $\mathbf{c}_i$ denotes the color feature, and $\mathbf{f}_i^s$ denotes the learnable semantic feature. The semantic feature enables the same Gaussian map to render RGB, depth, and semantic predictions. Given the current camera pose T_t, the differentiable Gaussian renderer produces RGB, depth, opacity, and semantic feature renderings:

$$\hat{I}_t, \hat{D}_t, \hat{A}_t, \hat{F}_t^s = \mathcal{R}(\mathcal{G}, T_t). \tag{2}$$

Here, $\hat{I}_t$, $\hat{D}_t$, $\hat{A}_t$, and $\hat{F}_t^s$ denote the rendered RGB image, rendered depth map, rendered opacity map, and rendered semantic feature map, respectively. The rendered semantic feature is decoded into an initial semantic probability map:

$$\hat{P}_t = \mathrm{softmax}\left(\Phi_s(\hat{F}_t^s)\right), \tag{3}$$

where Φ_s denotes the semantic decoder. Concord-SLAM further derives a local normal map from the rendered depth:

$$\hat{N}_t = \Psi(\hat{D}_t, K), \tag{4}$$

where K denotes the camera intrinsics and $\Psi(\cdot)$ computes depth-derived normals by back-projection and local finite differences. Thus, semantic, depth, and normal outputs are coupled renderings of the same Gaussian map.

3.2 CRC: Cross-Render Concordance

CRC couples semantic, depth, and normal renderings by using rendered geometry as evidence to predict a residual correction for semantic logits. Given the rendered semantic feature $\hat{F}_t^s$, rendered depth $\hat{D}_t$, and depth-derived normal $\hat{N}_t$, CRC predicts the semantic logit residual as

$$\begin{aligned} \Delta Z_t &= H_\theta\left(\phi_s(\hat{F}_t^s), \phi_d(\hat{D}_t), \phi_n(\hat{N}_t)\right), \\ \bar{P}_t &= \mathrm{softmax}\left(\Phi_s(\hat{F}_t^s) + \Delta Z_t\right). \end{aligned} \tag{5}$$

Here, ϕ_s, ϕ_d, and ϕ_n denote lightweight encoders for semantic features, depth, and normals, respectively, and H_θ denotes a relation network. The residual ΔZ_t does not generate a new semantic prediction from scratch. Instead, it performs structure-guided correction on the semantic logits decoded from the Gaussian map, allowing depth and normal structures to participate in semantic optimization while keeping the semantic output tied to the map representation. CRC further adjusts semantic supervision according to local geometric reliability. Depth edges and normal variations are combined into a geometry boundary response:

$$B_t^g = \frac{1}{2}\left(\mathcal{E}_d(\hat{D}_t) + \mathcal{E}_n(\hat{N}_t)\right), \tag{6}$$

where $E_d(\cdot)$ and $E_n(\cdot)$ denote the first-order depth-gradient response and adjacent-normal variation response, respectively. Their implementation details are provided in Appendix B.10. The geometry reliability weight is then defined as

$$W_t^g = \exp(-\tau B_t^g), \tag{7}$$

where τ controls the decay rate of the reliability weight near geometric discontinuities. We use $\tau = 8.0$ for all scenes.

A small B_t^g indicates a geometrically smooth region, where the semantic observation is usually reliable. A large B_t^g indicates an object boundary, occlusion boundary, or depth discontinuity, where hard semantic labels may be spatially misaligned. Based on this weight, CRC constructs a soft semantic target:

$$\tilde{Y}_t = W_t^g \cdot \mathrm{onehot}(Y_t^*) + (1 - W_t^g) \cdot \mathrm{sg}(\bar{P}_t), \tag{8}$$

where $\mathrm{sg}(\cdot)$ denotes the stop-gradient operation. In geometrically stable regions, the target is closer to the semantic observation Y_t^*, providing stronger direct supervision. Near geometric discontinuities, the influence of potentially misaligned hard labels is reduced, and the structure-corrected prediction is used as a stop-gradient soft target. The CRC loss is defined as the geometry-aware semantic consensus loss:

$$\mathcal{L}_{\mathrm{crc}} = -\sum_{p \in \Omega_t} \tilde{Y}_t(p)^\top \log \bar{P}_t(p), \tag{9}$$

where Ω_t denotes the valid pixel set. Different from a standard semantic cross-entropy loss that directly supervises the initial prediction $\hat{P}_t$, $\mathcal{L}_{\mathrm{crc}}$ acts on the structure-corrected prediction $\bar{P}_t$ and uses a geometry-aware soft target. In this way, CRC makes semantic optimization aware of local geometric structure, reducing semantic boundary misalignment and local category mixing.

3.3 BNS: Boundary-Native Splatting

CRC improves semantic consistency during rendering and optimization, whereas BNS addresses boundary ambiguity at the Gaussian growth stage. Since 3D Gaussians are explicit spatial primitives, a primitive inserted across an object boundary may be forced to explain different semantic or geometric regions simultaneously. BNS mitigates this problem by placing additional primitives near under-covered semantic boundaries, so that different sides of a boundary can be represented more separately. Concord-SLAM first follows the common new-Gaussian insertion strategy in Gaussian SLAM to detect regions insufficiently represented by the current map. The depth residual map is defined as

$$R_t^d(p) = \left|\hat{D}_t(p) - D_t^*(p)\right|. \tag{10}$$

The base insertion region is defined as

$$M_{\mathrm{base}} = \{\hat{A}_t < \tau_\alpha\} \cup \left\{R_t^d > \kappa \cdot \mathrm{median}(R_t^d),\ \hat{D}_t > D_t^*\right\}, \quad \kappa = 50. \tag{11}$$

Here, τ_α is the opacity threshold, and $\kappa \cdot \mathrm{median}(R_t^d)$ is the adaptive depth residual threshold. The opacity condition detects under-covered regions, while the depth condition detects missing foreground geometry. When $\hat{D}_t > D_t^*$, the rendered surface lies behind the observed surface, indicating that the current map fails to explain the visible foreground structure. Compared with a fixed depth threshold, the adaptive residual threshold reduces sensitivity to scene scale and depth noise. However, M_{base} only detects where Gaussians are missing. For semantic mapping, boundary regions may also require

Table 1: Localization and rendering results on the Replica dataset. ATE RMSE and Depth L1 are measured in centimeters. Higher values are better for PSNR and SSIM, while lower values are better for ATE, Depth L1, and LPIPS. The best result is highlighted in **blue and bolded**, the second-best result is highlighted in green, and the third-best result is highlighted in sand.

Method	ATE RMSE ↓									Depth L1 ↓	PSNR ↑	SSIM ↑	LPIPS ↓
	R0	R1	R2	O0	O1	O2	O3	O4	Avg.				
Gaussian-SLAM	0.44	0.32	0.31	0.44	0.52	**0.23**	0.17	2.25	0.58	1.50	38.94	0.968	0.070
GS-SLAM	0.48	0.53	0.33	0.52	0.41	0.59	0.46	0.70	0.50	1.16	34.27	0.975	0.082
SplaTAM	0.31	0.40	0.29	0.47	0.27	0.29	0.32	0.55	0.36	0.53	34.11	0.970	0.100
SGS-SLAM	0.46	0.45	0.29	0.46	0.23	0.45	0.42	0.55	0.41	0.36	34.66	0.973	0.096
GS^{3}LAM	0.27	0.25	0.28	0.67	0.21	0.33	0.30	0.65	0.37	0.87	36.26	**0.989**	0.052
SemGauss-SLAM	**0.26**	0.42	**0.27**	0.34	**0.17**	0.32	0.36	**0.49**	0.33	0.50	35.03	0.982	0.062
GSFF-SLAM	0.34	0.29	0.33	0.35	0.24	0.25	**0.15**	0.54	0.31	0.72	38.67	0.974	**0.035**
Ours	0.27	**0.22**	**0.27**	**0.33**	0.24	0.32	0.25	0.50	**0.30**	**0.35**	**39.57**	**0.989**	0.060

more separable primitive representations, even when they are partially covered. If the two sides of a boundary are jointly explained by the same group of Gaussians, the semantic boundary can be smoothed out during optimization. Therefore, BNS further introduces a semantic-boundary- driven insertion strategy. Given the semantic observation Y_t^*, a pixel p is regarded as a semantic boundary pixel if one of its neighboring pixels has a different semantic label:

$$B_t^s(p) = \mathbf{1}\left[\exists q \in \mathcal{N}(p), Y_t^*(p) \neq Y_t^*(q)\right], \tag{12}$$

where $\mathcal{N}(p)$ denotes the neighborhood of p. To tolerate projection errors, depth noise, and semantic misalignment, BNS applies a small dilation to the semantic boundary map and inserts additional Gaussians only at under-covered boundary locations:

$$M_{\text{bns}} = \text{Dilate}(B_t^s, r) \cap \{\hat{A}_t < \tau_b\} \cap \{D_t^* > 0\}. \tag{13}$$

Here, r is the boundary dilation radius, and τ_b is the opacity threshold for the boundary region. In this paper, we set $r = 1$ and $\tau_b = 0.5$. Thus, BNS does not insert Gaussians at all semantic boundaries indiscriminately. It supplements primitives only where boundary neighborhoods remain insufficiently covered. The final insertion region is

$$M_{\text{add}} = M_{\text{base}} \cup M_{\text{bns}}. \tag{14}$$

Here, M_{base} follows the standard opacity- and depth-residual-based insertion rule, while M_{bns} specifically captures under-covered semantic-boundary neighborhoods. Therefore, BNS does not replace the conventional Gaussian growth strategy. Instead, it augments it with semantic-boundary-aware insertion so that boundary regions can receive additional separable primitives when their opacity coverage is still insufficient. For each pixel $p \in M_{\text{add}}$, Concord-SLAM back-projects the observed depth into 3D space and initializes the center of a new Gaussian:

$$\mathbf{x}_p = T_t^{-1}\pi^{-1}(p, D_t^*(p), K). \tag{15}$$

Here, $\pi^{-1}(\cdot)$ denotes the back-projection operation from pixel coordinates and depth to a 3D point in the camera coordinate system, and T_t^{-1} transforms the point into the world coordinate system. The color of the new Gaussian is initialized from the current RGB observation, and its scale is estimated according to the depth value and camera intrinsics. Its semantic feature is initialized as a learnable vector and optimized in subsequent mapping steps through semantic supervision and CRC constraints. BNS is not an additional differentiable loss term, but a semantic-boundary-aware Gaussian allocation strategy. Standard densification mainly considers whether a region is sufficiently covered, while BNS further considers whether semantic boundaries need more separable primitives. By inserting new Gaussians near under-covered semantic boundaries, BNS reduces the probability that a single Gaussian spans multiple semantic regions, thereby alleviating semantic mixing and boundary blurring. The online tracking-and-mapping pipeline and the complete optimization objective are provided in Appendix A.

Table 2: Localization results on the ScanNet dataset. ATE RMSE is measured in centimeters, and lower values indicate better performance. The best result is highlighted in **blue and bolded**, the second-best result is highlighted in green, and the third-best result is highlighted in sand.

Method	0000	0059	0106	0169	0181	0207	Avg.
SGS-SLAM	11.15	9.54	**10.43**	10.70	11.28	6.11	**9.87**
SemGauss-SLAM	11.87	7.97	14.60	**8.70**	**9.78**	8.97	10.32
GSFF-SLAM	**9.60**	7.80	16.80	9.60	21.90	**5.60**	11.88
Ours	10.77	**7.53**	12.75	10.42	11.60	7.94	10.17

Table 3: Semantic mapping and ablation results on the Replica dataset, evaluated by mIoU (%). Ours* denotes the variant without Boundary-Native Splatting (BNS), and Ours** denotes the variant without Cross-Render Concordance (CRC). Higher values indicate better performance. The best result is highlighted in **blue and bolded**, the second-best result is highlighted in green, and the third-best result is highlighted in sand.

Method	R0	R1	R2	O0	O1	O2	O3	O4	Avg.
SGS-SLAM	92.95	92.91	92.10	92.90	91.45	92.67	92.38	91.93	92.41
GS^{3}LAM	96.83	96.68	96.40	96.61	96.01	95.67	95.12	96.74	96.26
SemGauss-SLAM	96.30	95.82	96.51	96.72	96.55	96.48	96.36	95.84	96.32
GSFF-SLAM	95.20	96.90	95.98	95.12	93.44	94.96	94.74	93.97	95.04
Ours	**98.35**	**98.88**	**98.54**	**98.27**	**98.40**	**98.06**	**97.51**	**97.15**	**98.15**
Ours*(w/o BNS)	97.69	97.64	97.50	96.31	96.78	96.12	96.55	96.48	96.88
Ours**(w/o CRC)	97.54	97.41	97.74	97.64	96.70	96.47	97.11	97.04	97.21

4 Experiments

4.1 Experimental Setup

We evaluate Concord-SLAM on two standard indoor RGB-D datasets, Replica [Straub et al., 2019] and ScanNet [Dai et al., 2017]. Replica is used for evaluating localization, reconstruction, rendering, and semantic mapping under synthetic indoor scenes, while ScanNet is used to further test localization robustness in real-world RGB-D sequences. Following common protocols, we use eight Replica scenes, including Room0, Room1, Room2, Office0, Office1, Office2, Office3, and Office4, denoted as R0, R1, R2, O0, O1, O2, O3, and O4. For ScanNet, we evaluate on six sequences, including 0000, 0059, 0106, 0169, 0181, and 0207. We compare Concord-SLAM with representative Gaussian SLAM and semantic Gaussian SLAM methods, including Gaussian-SLAM [Matsuki et al., 2024], GS-SLAM [Yan et al., 2024], SplaTAM [Keetha et al., 2024], SGS-SLAM [Li et al., 2024c], GS^{3}LAM [Li et al., 2024b], SemGauss-SLAM [Zhu et al., 2025], and GSFF-SLAM [Lu et al., 2026]. For localization, we report ATE RMSE in centimeters. For reconstruction and rendering, we report Depth L1, PSNR, SSIM, and LPIPS. For semantic mapping, we use mIoU. For runtime efficiency, we report Tracking FPS, Mapping FPS, and overall SLAM FPS. All experiments are conducted in an online RGB-D semantic SLAM setting. The system processes frames sequentially without using future-frame information. The first frame initializes the Gaussian map and defines the world reference frame. Subsequent frames are processed by alternating camera tracking and map optimization. Unless otherwise specified, results reported for existing methods are taken from the original papers when available, and missing entries are reproduced under the same experimental setting. Detailed dataset descriptions, metric definitions, baseline configurations, and implementation settings are provided in Appendix B.

4.2 Experimental Results

4.2.1 Localization Performance

As shown in Table 1 and Table 2, localization accuracy reflects whether the reconstructed map can provide stable constraints for camera tracking. On Replica, the proposed method achieves the best average ATE of 0.30, outperforming GSFF-SLAM with 0.31, SemGauss-SLAM with 0.33, SplaTAM with 0.36, and GS^{3}LAM with 0.37. Although it does not obtain the lowest error in every scene, its

Table 4: Runtime efficiency comparison in terms of Tracking FPS, Mapping FPS, and overall SLAM FPS. Higher values indicate better efficiency. The best result is highlighted in **blue and bolded**, the second-best result is highlighted in green, and the third-best result is highlighted in sand.

Method	Tracking FPS ↑	Mapping FPS ↑	SLAM FPS ↑
SGS-SLAM	5.27	3.52	2.11
GS^{3}LAM	2.77	1.75	1.07
GSFF-SLAM	3.77	2.78	1.60
Ours	**6.49**	**4.44**	**2.64**

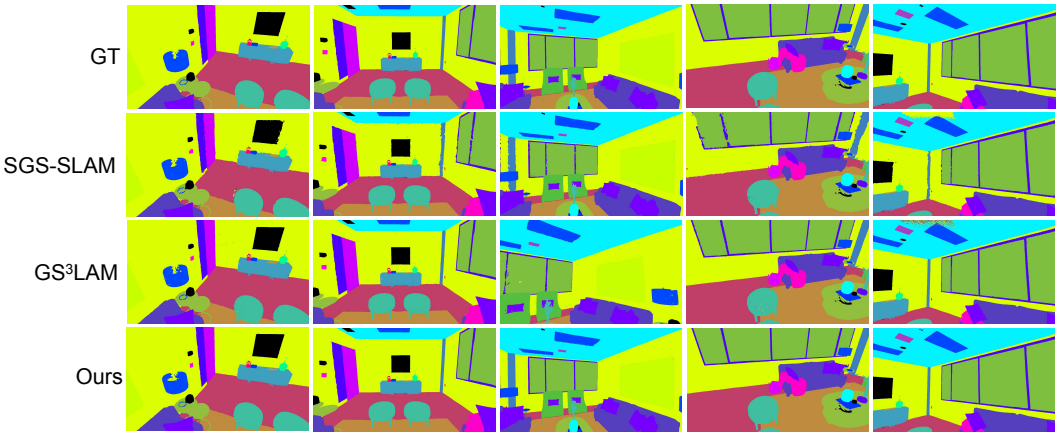


Figure 2: Qualitative comparison of semantic mapping results. Each column corresponds to a different test scene, and each row shows the semantic map produced by GT, SGS-SLAM, GS^{3}LAM, and the proposed method, respectively. The proposed method produces semantic maps closer to the ground truth, especially around object boundaries, contact regions, and fine structures.

best average result indicates stronger cross-scene stability and shows that the constructed semantic Gaussian map can support reliable tracking across different indoor environments. On ScanNet, the proposed method achieves an average ATE of 10.17, outperforming SemGauss-SLAM with 10.32 and GSFF-SLAM with 11.88, while being slightly inferior to SGS-SLAM with 9.87. This suggests that the method remains competitive on real-world RGB-D sequences, where sensor noise, missing depth values, complex motion, and trajectory drift are more prominent. Overall, the results show that the proposed semantic consistency modeling does not weaken camera tracking and can maintain robust localization performance.

4.2.2 Reconstruction Performance

Table 1 also reports reconstruction and rendering quality on Replica. The proposed method achieves the best Depth L1 of 0.35, outperforming SGS-SLAM with 0.36, SemGauss-SLAM with 0.50, SplaTAM with 0.53, and GS^{3}LAM with 0.87. This indicates that the proposed method renders more accurate depth maps and improves geometric consistency in the Gaussian representation. For appearance rendering, the proposed method obtains the highest PSNR of 39.57 and an SSIM of 0.989, tying with GS^{3}LAM for the best result. These results show that the method improves geometry and semantics without sacrificing RGB reconstruction quality. Its LPIPS is 0.060, which is better than SGS-SLAM and SemGauss-SLAM, but worse than GSFF-SLAM with 0.035 and GS^{3}LAM with 0.052. This indicates that the method still has room for improvement in perceptual similarity, especially for local textures and high-frequency appearance details.

4.2.3 Semantic Mapping Performance

Figure 2 shows that the proposed method produces semantic maps closer to the ground truth than SGS-SLAM and GS^{3}LAM. It better preserves category consistency in large planar regions and yields cleaner boundaries around objects, furniture edges, and adjacent semantic regions, indicating reduced

boundary ambiguity and cross-category diffusion. The quantitative semantic mapping results are reported in Table 3. The proposed method achieves the best semantic performance in all tested Replica scenes, with an average mIoU of 98.15. It outperforms SemGauss-SLAM with 96.32, GS^{3}LAM with 96.26, GSFF-SLAM with 95.04, and SGS-SLAM with 92.41. Compared with the closest competitor SemGauss-SLAM, it improves mIoU by 1.83 percentage points; compared with GSFF-SLAM and SGS-SLAM, the improvements are 3.11 and 5.74 percentage points, respectively. These gains mainly come from the complementary effects of BNS and CRC. BNS inserts new Gaussians near under-covered semantic-boundary neighborhoods, reducing the chance that one primitive spans different categories. CRC further uses depth and normal information to regularize semantic prediction, encouraging semantic boundaries to agree with geometric structures. Therefore, the semantic improvement is not only due to stronger semantic supervision, but also to the joint consistency modeling among semantics, depth, and normals.

4.2.4 Runtime Efficiency

Runtime efficiency is important for online dense semantic SLAM. As reported in Table 4, the proposed method achieves Tracking FPS, Mapping FPS, and overall SLAM FPS of 6.49, 4.44, and 2.64, respectively, all of which are the highest among the compared methods. Compared with SGS-SLAM, the proposed method improves Tracking FPS from 5.27 to 6.49, Mapping FPS from 3.52 to 4.44, and overall SLAM FPS from 2.11 to 2.64, corresponding to relative improvements of 23.1%, 26.1%, and 25.1%. These results show that the additional computations introduced by CRC and BNS remain limited. CRC performs lightweight residual correction on semantic logits using rendered depth and depth-derived normals as structural evidence, while BNS modifies the Gaussian growth mask only at under-covered boundary regions. Neither component requires global scene-level matching or a separate post-processing stage. As a result, Concord-SLAM preserves high online efficiency while improving semantic boundary representation and cross-render structural consistency.

4.2.5 Ablation Study

We conduct ablation experiments on Replica to evaluate the individual contributions of Boundary-Native Splatting (BNS) and Cross-Render Concordance (CRC). As shown in Table 3, Ours* denotes the variant without BNS, and Ours** denotes the variant without CRC. The full model achieves the best average semantic score of 98.15, outperforming both ablated variants. **Effect of BNS.** When BNS is removed, the average semantic score drops from 98.15 to 96.88, with a decrease of 1.27 percentage points. The degradation is particularly clear in scenes O0, O1, and O2, where the scores decrease from 98.27, 98.40, and 98.06 to 96.31, 96.78, and 96.12, respectively. These results indicate that BNS plays an important role in semantic Gaussian growth. By inserting new Gaussians around under-covered semantic-boundary neighborhoods, BNS improves boundary coverage and reduces cross-boundary semantic mixing. Without this boundary-aware growth strategy, Gaussians are more likely to spread across adjacent regions with different semantic labels, leading to blurred boundaries and less accurate semantic maps. **Effect of CRC.** When CRC is removed, the average semantic score decreases from 98.15 to 97.21, with a drop of 0.94 percentage points. This shows that CRC also contributes consistently to semantic mapping performance. Without CRC, semantic optimization relies more directly on semantic supervision and lacks explicit structural feedback from rendered depth and normal cues. As a result, inconsistent predictions around object boundaries are less effectively corrected. CRC mitigates this problem by enforcing cross-render consistency among semantic, depth, and normal outputs, allowing geometric structure to provide useful guidance for semantic refinement.

5 Conclusion

This paper presents Concord-SLAM, a self-consistent semantic Gaussian SLAM framework for dense semantic mapping. The key idea is to exploit the consistency among semantic, depth, and normal renderings from the same Gaussian field. To this end, Concord-SLAM introduces Cross-Render Concordance (CRC) to use cross-render structural inconsistencies as refinement signals, and Boundary-Native Splatting (BNS) to improve Gaussian growth around under-covered semantic-boundary regions. Experiments on Replica and ScanNet show that Concord-SLAM improves reconstruction, rendering, and semantic mapping quality, while maintaining competitive tracking performance., confirming the effectiveness of CRC and BNS.

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A SLAM System Framework and Optimization Details

Concord-SLAM follows an online tracking-and-mapping framework. Given an RGB-D semantic sequence $\{I_t^*, D_t^*, Y_t^*\}_{t=1}^T$, the system incrementally estimates the camera trajectory and constructs a persistent semantic Gaussian map $\mathcal{G}$. Each Gaussian primitive stores its position, scale, rotation, opacity, color feature, and learnable semantic feature. For each incoming frame, Concord-SLAM first estimates the current camera pose with the Gaussian map fixed, and then updates the semantic Gaussian map using the current observation together with historical keyframes.

A.1 Initialization

For the first frame, Concord-SLAM initializes the world coordinate system using the camera coordinate system of that frame. Each valid depth pixel is back-projected into a 3D point and used to initialize a Gaussian primitive. The color of each Gaussian is initialized from the corresponding RGB observation, and the scale is estimated according to the depth value and camera intrinsics:

$$s_p \propto \frac{D_0^*(p)}{(f_x + f_y)/2}. \tag{16}$$

Here, $D_0^*(p)$ denotes the observed depth of pixel p in the first frame, and f_x and f_y denote the focal lengths of the camera. A learnable semantic feature is also initialized for each Gaussian primitive, so that the map can subsequently render semantic predictions in addition to RGB and depth.

A.2 Tracking

For each incoming frame t, Concord-SLAM first obtains an initial camera pose estimate using a constant-velocity motion model. Then, with the semantic Gaussian map $\mathcal{G}$ fixed, the current camera pose is optimized by minimizing a tracking objective:

$$T_t^\star = \arg\min_{T_t} \mathcal{L}_{\text{track}}. \tag{17}$$

Given a candidate pose T_t, the differentiable Gaussian renderer produces RGB, depth, opacity, and semantic renderings from the current map. The tracking loss is defined as

$$\mathcal{L}_{\text{track}} = \lambda_I \mathcal{L}_{\text{rgb}} + \lambda_D \mathcal{L}_{\text{depth}} + \lambda_S \mathcal{L}_{\text{sem}}. \tag{18}$$

Here, $\mathcal{L}_{\text{rgb}}$, $\mathcal{L}_{\text{depth}}$, and $\mathcal{L}_{\text{sem}}$ denote RGB, depth, and semantic reprojection losses, respectively. During tracking, the Gaussian map parameters are kept fixed, and only the camera pose of the current frame is optimized. To improve tracking robustness, the loss is computed only over valid depth regions and reliable mapped regions. Specifically, regions with insufficient opacity coverage are excluded because they are not yet well represented by the current Gaussian map. Pixels near semantic boundaries can also be ignored during tracking, since they are more likely to be affected by occlusion, depth discontinuities, and semantic-label misalignment. During pose optimization, the pose with the lowest tracking loss is recorded as the final pose estimate for the current frame.

A.3 Gaussian Growth and Boundary-Native Splatting

After estimating the current camera pose, Concord-SLAM updates the semantic Gaussian map. Before continuous optimization, the system determines where new Gaussian primitives should be inserted. The base growth strategy follows common Gaussian SLAM practice and detects regions insufficiently represented by the current map according to opacity coverage and depth residuals. Low rendered opacity indicates that the region is not sufficiently covered by existing Gaussians. A large depth residual, especially when the rendered depth lies behind the observed depth, indicates that the current map fails to explain visible foreground geometry. Boundary-Native Splatting (BNS) further extends this strategy by considering semantic boundaries. Instead of inserting new Gaussians only in globally under-covered regions, BNS gives additional attention to under-covered neighborhoods around semantic boundaries. This design encourages different sides of a semantic boundary to be represented by more separable Gaussian primitives. Therefore, BNS reduces the chance that a single Gaussian primitive spans multiple semantic regions, which helps alleviate boundary blurring and semantic mixing from the map growth stage. The final insertion mask is obtained by combining the base insertion region and the BNS boundary insertion region. For each selected pixel, the system back-projects the pixel into 3D using the observed depth and initializes a new Gaussian primitive. Its position is initialized by back-projection, its color is initialized from the RGB observation, and its semantic feature is learned during subsequent mapping optimization.

A.4 Mapping Optimization

After Gaussian insertion, Concord-SLAM optimizes the semantic Gaussian map using the current frame and historical keyframes. The optimized variables include Gaussian centers, scales, rotations, opacities, color features, and semantic features:

$$\{\boldsymbol{\mu}_i, \mathbf{s}_i, \mathbf{q}_i, \alpha_i, \mathbf{c}_i, \mathbf{f}_i^s\}_{i=1}^{N}. \tag{19}$$

The mapping stage jointly optimizes the Gaussian map, the semantic decoder, and the CRC module:

$$\mathcal{G}^\star, \Phi_s^\star, \Theta_{\text{crc}}^\star = \arg\min_{\mathcal{G}, \Phi_s, \Theta_{\text{crc}}} \mathcal{L}_{\text{map}}. \tag{20}$$

Here, Φ_s denotes the semantic decoder, and Θ_{crc} denotes the parameters of the Cross-Render Concordance module. The mapping objective is defined as

$$\mathcal{L}_{\text{map}} = \lambda_I \mathcal{L}_{\text{rgb}} + \lambda_D \mathcal{L}_{\text{depth}} + \lambda_S \mathcal{L}_{\text{sem}} + \lambda_C \mathcal{L}_{\text{crc}} + \lambda_R \mathcal{L}_{\text{reg}}. \tag{21}$$

Here, $\mathcal{L}_{\text{rgb}}$ denotes the RGB reconstruction loss, $\mathcal{L}_{\text{depth}}$ denotes the depth reconstruction loss, $\mathcal{L}_{\text{sem}}$ denotes the semantic supervision loss, $\mathcal{L}_{\text{crc}}$ denotes the Cross-Render Concordance loss, and $\mathcal{L}_{\text{reg}}$ denotes regularization terms for stabilizing Gaussian geometry, opacity, and scale. The RGB loss combines pixel-wise reconstruction and structural similarity terms. The depth loss is computed over valid depth regions. The semantic loss is computed between the semantic labels and the logits decoded from the rendered semantic features. The CRC loss further constrains semantic prediction using geometric evidence from rendered depth and depth-derived normals. In this way, the semantic, depth, and normal renderings from the same Gaussian field are encouraged to provide structurally consistent explanations of the scene.

A.5 Roles of CRC and BNS

CRC and BNS affect Concord-SLAM in different but complementary ways. CRC acts during differentiable optimization. It mainly optimizes the semantic features of Gaussian primitives, the semantic decoder, and the cross-render residual network through the mapping loss. By using rendered depth and depth-derived normals as structural evidence, CRC improves semantic prediction and encourages semantic boundaries to be consistent with geometric discontinuities. BNS acts during map growth. It does not introduce an additional differentiable loss term. Instead, it changes the spatial allocation of newly inserted Gaussian primitives by placing additional Gaussians near under-covered semantic-boundary regions. This makes the explicit Gaussian representation more boundary-aware before subsequent optimization. Therefore, BNS improves boundary separability at the representation-growth level, while CRC improves cross-render structural consistency at the rendering-optimization level. Together, they enable Concord-SLAM to construct a dense semantic Gaussian map with clearer boundaries, more stable semantics, and more coherent geometry.

A.6 Keyframe Maintenance

Concord-SLAM stores keyframes at fixed frame intervals. Each keyframe contains the RGB image, depth map, semantic labels, and the estimated camera pose. During mapping optimization, the system alternates between the current frame and randomly sampled historical keyframes. This replay-based strategy allows the map to absorb new observations while preserving consistency with previous views. To control map size and avoid redundant primitives, the system periodically prunes low-opacity or abnormal Gaussians and regularizes Gaussian scales. Overall, Concord-SLAM follows an online pipeline consisting of pose estimation, boundary-aware Gaussian growth, map optimization, and keyframe maintenance. Tracking aligns the current frame to the existing Gaussian map, while mapping constructs a structurally self-consistent and boundary-aware semantic Gaussian field through BNS and CRC.

B Additional Experimental Details

B.1 Datasets and Evaluation Protocol

We evaluate Concord-SLAM on two indoor RGB-D datasets, Replica [Straub et al., 2019] and ScanNet [Dai et al., 2017]. Replica is a synthetic indoor dataset with high-quality RGB images, depth maps, semantic annotations, and ground-truth camera trajectories. It provides clean geometry and accurate camera poses, and is therefore suitable for evaluating dense SLAM systems in terms of localization, reconstruction, rendering, and semantic mapping. Following common practice in dense RGB-D SLAM and Gaussian SLAM, we conduct experiments on eight representative Replica scenes: Room0, Room1, Room2, Office0, Office1, Office2, Office3, and Office4. These scenes are denoted as R0, R1, R2, O0, O1, O2, O3, and O4, respectively. ScanNet is a real-world indoor RGB-D dataset. Compared with Replica, ScanNet contains stronger sensor noise, missing depth values, occlusions, motion blur, and less accurate camera trajectories. Therefore, it is used to evaluate the robustness and generalization ability of the proposed method in more realistic indoor environments. Following the evaluation setting of related Gaussian SLAM methods, we select six ScanNet sequences, including 0000, 0059, 0106, 0169, 0181, and 0207. We report the absolute trajectory error of each sequence and the average result over all selected sequences. All experiments are conducted under an online RGB-D semantic SLAM setting. The system receives RGB-D images and semantic labels frame by frame, without using future-frame information. The first frame is used to initialize the semantic Gaussian map, and its camera coordinate system is used as the world reference frame. For each subsequent frame, the camera pose is first estimated during tracking, and the semantic Gaussian map is then updated during mapping. Final evaluation is performed every 5 frames.

B.2 Evaluation Metrics

We quantitatively evaluate Concord-SLAM from four aspects: pose estimation, geometric reconstruction, RGB rendering, and semantic mapping. For pose estimation, we report the absolute trajectory error root mean square error, denoted as ATE RMSE. The estimated trajectory is first aligned with the ground-truth trajectory, and the translational error is then computed in centimeters. Lower ATE RMSE indicates more accurate camera tracking. For geometric reconstruction, we report Depth L1

Table 5: Key implementation hyperparameters of Concord-SLAM.

Hyperparameter	Replica	ScanNet
Geometry reliability temperature τ	8.0	8.0
RGB loss weight λ_I	0.5	0.5
Depth loss weight λ_D	1.0	1.0
Semantic loss weight λ_S	0.008	0.008
CRC soft semantic-consensus weight	0.015	0.015
Semantic-geometry normal consistency weight	0.008	0.008
Large-Gaussian regularization weight	0.02	0.02
Small-Gaussian regularization weight	0.002	0.002
Base opacity threshold τ_α	0.1	0.5
Depth residual multiplier κ	50	50
Boundary dilation radius r	1	1
Boundary opacity threshold τ_b	0.5	0.5
Tracking iterations	40	200
Mapping iterations per frame	60	60
First-frame mapping iterations	1000	1000
Keyframe interval	5	5
Configured mapping window size	24	10

Table 6: Architecture of the CRC module. Conv$k\times k(a, b)$ denotes a convolutional layer with kernel size $k\times k$, input channels a, and output channels b.

Component	Input	Layers	Output
Semantic encoder ϕ_s	$\hat{F}_t^s$, 16 ch	Conv3×3(16,64)+ReLU; Conv3×3(64,64)+ReLU	64 ch
Depth encoder ϕ_d	$\hat{D}_t$, 1 ch	Conv3×3(1,64)+ReLU; Conv3×3(64,64)+ReLU	64 ch
Normal encoder ϕ_n	$\hat{N}_t$, 3 ch	Conv3×3(3,64)+ReLU; Conv3×3(64,64)+ReLU	64 ch
Pairwise fusion	$[\phi_s, \phi_d]$, $[\phi_s, \phi_n]$, $[\phi_d, \phi_n]$	Conv3×3(128,64)+ReLU for each pair	Three 64-ch maps
Relation block H_θ	Concatenated pairwise features, 192 ch	Conv3×3(192,64)+ReLU; Conv3×3(64,64)+ReLU	64 ch
Semantic delta head	64 ch	Conv1×1(64,256)	ΔZ_t, 256 ch

over valid depth pixels. Depth L1 measures the average absolute difference between rendered depth and observed or ground-truth depth. It is reported in centimeters, and lower values indicate more accurate geometry. For RGB rendering quality, we report PSNR, SSIM, and LPIPS. Higher PSNR and SSIM indicate better pixel-level and structural reconstruction quality, while lower LPIPS indicates better perceptual similarity. For semantic mapping, we use mean Intersection over Union, namely mIoU. It measures the overlap between predicted semantic regions and ground-truth semantic regions across categories. Higher mIoU indicates better semantic reconstruction. For runtime efficiency, we report Tracking FPS, Mapping FPS, and overall SLAM FPS. Tracking FPS measures the speed of pose estimation, Mapping FPS measures the speed of map update, and SLAM FPS reflects the overall online system throughput. Higher FPS indicates better runtime efficiency.

B.3 Baseline Methods

We compare Concord-SLAM with representative Gaussian SLAM and semantic Gaussian SLAM methods, including Gaussian-SLAM [Matsuki et al., 2024], GS-SLAM [Yan et al., 2024], SplaTAM [Keetha et al., 2024], SGS-SLAM [Li et al., 2024c], GS^{3}LAM [Li et al., 2024b], SemGauss-SLAM [Zhu et al., 2025], and GSFF-SLAM [Lu et al., 2026]. Gaussian-SLAM, GS-SLAM, and SplaTAM are mainly used to evaluate localization, reconstruction, and RGB rendering performance. These methods adopt 3D Gaussian Splatting or related splat-based representations for dense RGB-D SLAM, and are representative baselines for Gaussian-map-based tracking and mapping. SGS-SLAM, GS^{3}LAM, SemGauss-SLAM, and GSFF-SLAM further introduce semantic representations or semantic optimization mechanisms into Gaussian SLAM. Therefore, they are used as the main semantic Gaussian SLAM baselines for semantic mapping comparison. For runtime efficiency, we compare Concord-SLAM with SGS-SLAM, GS^{3}LAM, and GSFF-SLAM using Tracking FPS, Mapping FPS, and overall SLAM FPS. For results already reported in the original papers, we preferentially adopt the published values. For missing scenes or missing metrics, we reproduce and supplement the results under the same experimental setting as the corresponding original method.

B.4 Implementation Settings

All experiments were conducted on a workstation equipped with a single NVIDIA GeForce RTX 5090 GPU with 32 GB of GDDR7 memory. Each sequence was processed on one GPU without multi-GPU parallelization. The runtime efficiency reported in Table 4 was measured under the same hardware setting. Unless otherwise specified, all baseline reproductions and ablation experiments were conducted using the same workstation to ensure a fair comparison. The scene is represented using isotropic 3D Gaussian primitives. Each Gaussian stores its position, color, opacity, scale, rotation, and a 16-dimensional semantic feature. The semantic feature is decoded into semantic logits through a lightweight convolutional semantic decoder. The decoder outputs predictions over 256 semantic categories. We use the Adam optimizer for camera tracking and map optimization. Different learning rates are used for different groups of parameters, including camera poses, Gaussian geometric parameters, color parameters, opacity, scale, rotation, and semantic features. This follows the common practice in Gaussian SLAM, where different Gaussian attributes have different numerical ranges and optimization sensitivities. On Replica, tracking is optimized for 40 iterations per frame. The first mapping frame is optimized for 1000 iterations, while each subsequent mapping frame is optimized for 60 iterations. The keyframe interval is set to 5, and the mapping window size is set to 24. On ScanNet, due to stronger sensor noise, missing depth values, and more complex camera motion, tracking is optimized for 200 iterations per frame. The mapping configuration remains the same as that on Replica, while the mapping window size is set to 10. All sequences are processed with a frame stride of 1. Final evaluation is performed every 5 frames. Table 5 summarizes the key hyperparameters used in the final implementation. Unless otherwise specified, the loss weights are shared by Replica and ScanNet. The compact mapping objective is implemented with separate weights for RGB reconstruction, depth reconstruction, semantic supervision, cross-render semantic-geometric consistency, and Gaussian regularization.

B.5 Tracking Details

During tracking, Concord-SLAM mainly optimizes the camera pose of the current frame while keeping the semantic Gaussian map fixed. The pose of the current frame is initialized using a constant-velocity motion model. Given the initialized pose, the differentiable Gaussian renderer produces RGB, depth, opacity, and semantic predictions from the current Gaussian map. The tracking loss consists of RGB reconstruction loss, depth loss, and a weak semantic constraint. The RGB loss measures the difference between the rendered RGB image and the input RGB image. The depth loss measures the difference between rendered depth and observed depth over valid depth pixels. The weak semantic constraint provides additional semantic alignment between the rendered semantic prediction and the input semantic label. To improve tracking robustness, the loss is computed over high-confidence mapped regions. Specifically, regions with low rendered opacity are excluded because they are not sufficiently covered by the current Gaussian map. Pixels near semantic boundaries are also treated cautiously, since they are more likely to be affected by occlusion boundaries, depth discontinuities, and semantic-label misalignment. This design reduces the influence of unmapped regions and ambiguous boundary pixels on pose estimation. During tracking optimization, the system records the pose with the lowest tracking loss and uses it as the final estimated pose of the current frame.

B.6 Mapping Details

During mapping, Concord-SLAM optimizes the geometry, color, opacity, and semantic features of the Gaussian primitives under the estimated camera pose. The mapping objective includes RGB reconstruction, depth reconstruction, semantic supervision, Cross-Render Concordance, and Gaussian regularization. The RGB loss consists of an L1 reconstruction term and a structural similarity term. The depth loss is computed as the L1 error over valid depth regions. The semantic loss is implemented as pixel-wise cross-entropy between the decoded semantic logits and the semantic labels. To reduce forgetting and improve global map consistency, Concord-SLAM optimizes both the current frame and historical keyframes during mapping. At fixed iteration intervals, the current frame is used for optimization. For the remaining iterations, frames are randomly sampled from the keyframe set. This keyframe-replay strategy allows the map to absorb new observations while preserving consistency with previous views. The Cross-Render Concordance module is activated during mapping after the map becomes relatively stable. It fuses rendered semantic features, rendered depth, and

normals derived from rendered depth, and predicts a correction term for semantic logits. CRC is not used during tracking. During final evaluation, the corrected semantic output is used for semantic prediction, while the depth metric is computed from the depth rendered by the base Gaussian map, ensuring comparability with related methods. For semantic boundary regions, Concord-SLAM adopts Boundary-Native Splatting. This strategy inserts new Gaussians near semantic boundaries where opacity coverage is insufficient. By allocating additional primitives around under-covered semantic-boundary regions, BNS improves the representation of object boundaries and reduces cross-category semantic mixing.

B.7 Keyframe and Map Maintenance

Concord-SLAM saves keyframes at fixed frame intervals. Each keyframe stores the RGB image, depth map, semantic labels, and the estimated camera pose. During mapping, the system alternates between the current frame and randomly sampled keyframes to maintain temporal consistency across the reconstructed map. To control map size and reduce redundancy, Concord-SLAM periodically prunes abnormal Gaussian primitives. Gaussians with very low opacity or unstable scale values are removed. Gaussian scales are also regularized during optimization to avoid degenerate primitives. These maintenance operations help stabilize long-sequence mapping and improve runtime efficiency.

B.8 CRC Architecture

CRC is implemented as a lightweight convolutional multimodal coupling network. The network takes the rendered semantic feature map, rendered depth map, and depth-derived normal map as inputs. Three modality-specific encoders first map them into 64-channel embeddings. Pairwise semantic-depth, semantic-normal, and depth-normal features are then fused and passed to a relation block, which predicts a semantic logit residual ΔZ_t. No BatchNorm, LayerNorm, or GroupNorm is used. The activation function is ReLU. CRC is enabled during mapping but disabled during tracking. During evaluation, the CRC-corrected semantic logits are used for semantic prediction, while the rendered depth is kept from the base Gaussian map.

B.9 BNS Implementation

The base insertion mask consists of two conditions: low opacity and large missing-foreground depth residual. Specifically, the opacity condition is $\hat{A}_t < \tau_\alpha$, where $\tau_\alpha = 0.1$ on Replica and $\tau_\alpha = 0.5$ on ScanNet. The depth-residual condition is activated when $\hat{D}_t(p) > D_t^*(p)$ and $|\hat{D}_t(p) - D_t^*(p)| > 50 \cdot \mathrm{median}(R_t^d)$. The semantic boundary map is computed from the semantic observation Y_t^* using 4-neighbor label differences, and both sides of a semantic transition are marked as boundary pixels. A max-pooling dilation with radius $r = 1$ is then applied. The BNS mask is defined as the dilated semantic boundary intersected with $\hat{A}_t < \tau_b$ and $D_t^* > 0$, where $\tau_b = 0.5$. The final insertion mask is the union of the base mask and the BNS mask. Depth edges and normal edges are not directly used in the BNS mask; they contribute to CRC instead.

B.10 Depth-derived normals and geometric boundary response

Given a rendered depth map $\hat{D}_t$ and camera intrinsics $K = (f_x, f_y, c_x, c_y)$, we back-project each pixel into a 3D point map. Local tangent directions are computed by centered finite differences on the point map, and the normal is obtained by the cross product of the two tangent directions followed by ℓ_2 normalization with $\epsilon = 10^{-6}$. The depth edge response E_d is computed by first-order forward differences:

$$E_d(\hat{D}_t)(p) = \sqrt{\left(\nabla_x \hat{D}_t(p)\right)^2 + \left(\nabla_y \hat{D}_t(p)\right)^2 + \epsilon}, \tag{22}$$

where ∇_x and ∇_y denote horizontal and vertical forward differences with replicate padding, and $\epsilon = 10^{-8}$ is used for numerical stability. The normal variation response E_n measures local angular changes between adjacent unit normals:

$$E_n(\hat{N}_t)(p) = \frac{1}{2}\left(1 - |\langle \hat{n}_p, \hat{n}_{p+\mathbf{e}_x} \rangle| + 1 - \left|\langle \hat{n}_p, \hat{n}_{p+\mathbf{e}_y} \rangle\right|\right), \tag{23}$$

where $\hat{n}_p$ denotes the depth-derived normal at pixel p, and $\mathbf{e}_x$ and $\mathbf{e}_y$ denote horizontal and vertical neighboring offsets. The final geometric boundary response and reliability weight are

$$B_t^g = \frac{1}{2}\left(E_d(\hat{D}_t) + E_n(\hat{N}_t)\right), \qquad W_t^g = \exp(-8.0 B_t^g). \tag{24}$$

C Limitations

Although Concord-SLAM improves semantic Gaussian SLAM through cross-render consistency and boundary-native Gaussian growth, several limitations remain. First, the current framework relies on closed-set semantic predictions from an external semantic segmentation model. As a result, its semantic mapping ability is constrained by the predefined label space and the quality of the input semantic predictions. Extending Concord-SLAM to open-vocabulary semantic understanding would improve its applicability to more general real-world environments. Second, Concord-SLAM is mainly designed for static indoor scenes. Dynamic objects may introduce inconsistent observations across frames and affect both Gaussian optimization and semantic consistency modeling. Handling dynamic scenes remains an important direction for future work. Third, BNS introduces additional Gaussians around under-covered semantic-boundary regions. Although this improves boundary representation, it may increase memory usage and optimization cost in large-scale scenes. More efficient boundary-adaptive Gaussian management strategies are needed to balance mapping quality and computational efficiency. Finally, CRC assumes that semantic, depth, and normal renderings should be structurally consistent. This assumption is generally valid in static scenes, but errors in depth estimation, noisy normals, or inaccurate semantic predictions may still affect the reliability of the consistency signals. Improving the robustness of cross-render consistency under noisy observations is another promising direction.

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